EC313 Assignment 1: Who Paid the Carbon Tax at the Pump? (Answer Key)

Answer key

2026-10-21

Background

From 2019 to 2025, the federal government charged a carbon price on fuel in provinces that did not have a carbon pricing system of their own that met the federal standard. The charge was paid to the government by fuel distributors, not by drivers, and from April 2024 it was 17.61 cents per litre of gasoline. British Columbia had its own carbon tax at the same rate. Quebec prices carbon through a cap-and-trade system instead, with no separate charge per litre at the pump. Most of the revenue from the federal charge was returned to households as the Canada Carbon Rebate, a payment that did not depend on how much fuel a household bought.

On April 1, 2025, the federal charge was set to zero, and British Columbia ended its carbon tax on the same day. Nothing changed in Quebec.

Who bore the carbon tax at the pump? The theory of tax incidence says that the answer depends on the elasticities of supply and demand, not on who sends the money to the government. If sellers had passed the whole tax on to drivers, the price of gasoline should have fallen by the full amount of the tax when it was removed. If sellers had absorbed part of the tax themselves, the price should have fallen by less. In this assignment you will measure how much the price fell, and use that measurement to work out who bore the tax.

The data

The data come from Statistics Canada's monthly survey of retail gasoline prices (Table 18-10-0001-01). They are the average price of regular unleaded gasoline at self-service stations in 15 cities, for each month from October 2024 to September 2025: six months before the carbon tax was removed, and six months after.

Variable	Description
`city`	City
`province`	Province
`date`	The month, stored as a date (the first day of the month)
`month`	The month, as text, such as "2025-03" for March 2025
`gas`	Average price of regular gasoline, in cents per litre, including all taxes
`salestax`	The GST or HST rate charged on gasoline in the province (plus the QST in Quebec), as a proportion: 0.13 means 13%

The GST and HST are charged on the price of gasoline *including* the carbon charge, so removing the 17.61 cent charge removes 17.61 × (1 + `salestax`) cents from the price at the pump.

Winnipeg is left out, because Manitoba's own gas tax went back up by 12.5 cents in January 2025, in the middle of the period. Whitehorse and Yellowknife are left out too, because prices there come from few stations and often do not change for months at a time. In Halifax, the HST also fell from 15% to 14% on April 1, 2025, which lowered the price there by about one cent more.

Source: Statistics Canada, Table 18-10-0001-01, Monthly average retail prices for gasoline and fuel oil, by geography, October 2024 to September 2025. Reproduced and distributed on an "as is" basis with the permission of Statistics Canada.

Instructions

- Write your R code in the chunks provided. You can add more chunks if you need them.
- When a question tells you to store a result under a particular name, such as `q2` or `pt_naive`, use exactly that name. Those objects are marked by computer, and a different name counts as missing.
- Do not round the numbers you store. Round only when you display them.
- Tables and graphs must be presentable: use `modelsummary()`, `datasummary()` or `datasummary_df()` for tables, and give every graph readable axis labels.
- The hints under each question point to sections of the R tutorial, the Introduction to R and Quarto page on the course website. Each section has a worked example of the step the hint describes, using different data.
- Render the document after each question, not only at the end.
- Every written answer goes under **Answer:**, and what it must cover is set out just above, under **Comment on the results**.
- Each question is marked out of 5 from your PDF, using the rubric on the assignment page. The objects you are told to create are also checked by computer, for another 5 marks, so the assignment is out of 30.

Setup

Do not change this chunk or the next one.

```
library(tidyverse)
library(modelsummary)
```

```
gas <- read_csv("ec313assign1.csv")

glimpse(gas)
```

```
Rows: 180
Columns: 6
$ city     <chr> "Calgary", "Calgary", "Calgary", "Calgary", "Calgary", "Calga~
$ province <chr> "Alberta", "Alberta", "Alberta", "Alberta", "Alberta", "Alber~
$ date     <date> 2024-10-01, 2024-11-01, 2024-12-01, 2025-01-01, 2025-02-01, ~
$ month    <chr> "2024-10", "2024-11", "2024-12", "2025-01", "2025-02", "2025-~
$ gas      <dbl> 146.5, 142.4, 140.8, 149.2, 150.0, 147.9, 130.2, 131.2, 131.2~
$ salestax <dbl> 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0~
```

Question 1 (5 marks)

Create the following variables and add them to `gas`:

- `removed`: 1 if the city's carbon tax was removed on April 1, 2025 (every city outside Quebec), 0 for the two Quebec cities
- `post`: 1 for April 2025 and every month after it, 0 for the months before
- `carbon`: the carbon tax included in the price, in cents per litre, counting the sales tax charged on it. It is 17.61 $\times$ (1 + `salestax`) in cities where the tax was removed, in the months before April 2025. It is 0 in every other case: in those cities from April 2025 on, and in the Quebec cities in every month.

Then make a table showing the value of `carbon` in each of the 15 cities in March 2025, the last month before the tax was removed.

Hints.

- Create all three variables with `mutate()`, and save the result back into `gas`; otherwise the new variables are not kept. Inside one `mutate()`, a variable you have just created can be used in the lines below it. See "`mutate()` — create variables" in the R tutorial.
- For `removed`, a comparison such as `province == "Quebec"` gives `TRUE` or `FALSE` for every row, and `!=` means "is not equal to". `as.numeric()` turns `TRUE` and `FALSE` into 1 and 0. See "Picking out elements" and "Vectors have a type" in the R tutorial.

- For `post`, comparisons such as `>=` also work on dates. To compare `date` with April 1, 2025, write that date as `as.Date("2025-04-01")`. See "Dates" in the R tutorial.
- For `carbon`, `if_else(condition, a, b)` gives `a` in the rows where the condition is `TRUE` and `b` in the rows where it is `FALSE`. A condition can combine two others with `&` (and), such as one about `removed` and one about `post`. In R, multiplication is `*`, not $\times$. See "`mutate()` — create variables" in the R tutorial.
- For the table, use `datasummary(city ~ carbon * Mean, data = ...)`, with `data` set to the March 2025 rows of `gas` (use `filter()`). Each city has only one row in March, so its mean is just its March value. Give the column a clearer heading than "Mean". See "Tables of data" in the R tutorial.

```
gas <- gas |>
  mutate(
    removed = as.numeric(province != "Quebec"),
    post    = as.numeric(date >= as.Date("2025-04-01")),
    carbon  = if_else(removed == 1 & post == 0, 17.61 * (1 + salestax), 0)
  )
```

```
datasummary(city ~ carbon * Heading("Carbon tax (cents per litre)") * Mean,
            data = filter(gas, month == "2025-03"), fmt = 2)
```

city	Carbon tax (cents per litre)
Calgary	18.49
Charlottetown	20.25
Edmonton	18.49
Halifax	20.25
Montreal	0.00
Ottawa	19.90
Quebec City	0.00
Regina	18.49
Saint John	20.25
Saskatoon	18.49
St. John's	20.25
Thunder Bay	19.90
Toronto	19.90
Vancouver	18.49
Victoria	18.49

Comment on the results. Use the theory of tax incidence from class to predict what should have happened to the price of gasoline in Toronto when the tax was removed. By how much should the price have fallen if the supply of gasoline were perfectly elastic? By how much if it were perfectly inelastic? Which of the two do you expect to be closer to what happened, and why?

Answer:

The carbon tax in the price in Toronto was 19.90 cents a litre: the 17.61 cent charge plus the 13% HST charged on it.

If the supply of gasoline were perfectly elastic, sellers would supply any quantity at the same price, net of tax, and nothing at a lower one. They could not absorb any of the tax, so drivers would bear all of it, and the price at the pump should have fallen by the full 19.90 cents.

If supply were perfectly inelastic, sellers would supply the same quantity whatever the price, so the price drivers pay would be set by demand at that quantity, which the tax does not change. The price at the pump would not have changed at all. Sellers would have borne the whole tax, and removing it would have raised the price they receive by 19.90 cents.

I expect the price to have fallen by close to the full 19.90 cents, for two reasons. The supply of gasoline to one city is likely to be very elastic: refiners and wholesalers sell into a North American market, and if the price they could get in Toronto fell, they would send their gasoline somewhere else. And demand for gasoline is inelastic in the short run, because drivers cannot quickly change how far they drive or what car they own. The more elastic supply is relative to demand, the larger the share of the tax that drivers bear.

Question 2 (5 marks)

For each month, calculate the average price of gasoline in the cities where the carbon tax was removed, and the average in the two Quebec cities. That gives 24 averages: 12 months, for each of the two groups. Store the result in a data frame called `q2`, with the averages in a column called `avg_gas`.

Then make a line graph of the two averages over time, with the date on the horizontal axis, one line for each group, and a vertical line marking April 2025.

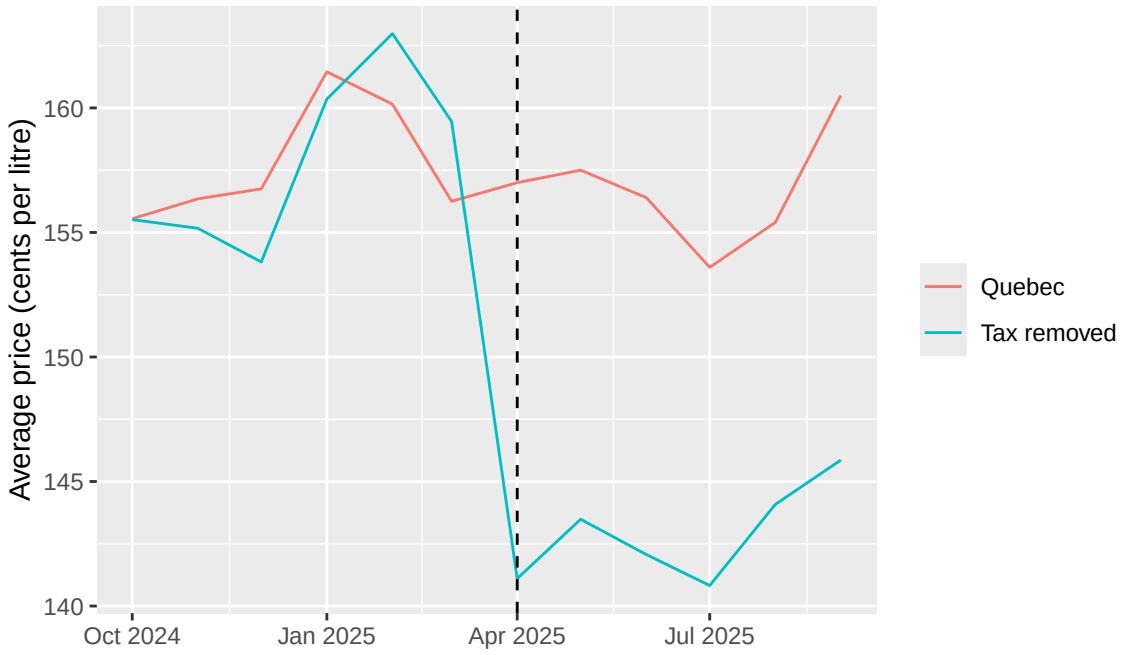
Hints.

- For the averages, use `group_by(date, removed)` and then `summarise()`. Grouping by both variables gives a separate average for each month in each group: the 24 rows. See "`group_by()` — do it separately for each group" in the R tutorial.
- You will need the averages for your comment. To see them, run the chunk (the green arrow at its top right), then type `q2` in the Console.

- For the graph, start with `ggplot(q2, aes(x = date, y = avg_gas))` and add `geom_line()`. See "Line graphs and groups" in the R tutorial, which has a graph like this one.
- Add `colour = factor(removed)` inside `aes()`. This gives each group its own line. Without `factor()`, R treats `removed` as a number rather than as two groups, and draws a single zigzag line.
- Add `geom_vline(xintercept = as.Date("2025-04-01"), linetype = "dashed")` to put in the vertical line at April 2025.
- Use `scale_colour_discrete()` to replace the 0 and 1 in the legend with words, giving the labels in the order 0, then 1, and use `labs()` to give the axes readable titles. See "Line graphs and groups" in the R tutorial.

```
q2 <- gas |>
  group_by(date, removed) |>
  summarise(avg_gas = mean(gas), .groups = "drop")
```

```
ggplot(q2, aes(x = date, y = avg_gas, colour = factor(removed))) +
  geom_line() +
  geom_vline(xintercept = as.Date("2025-04-01"), linetype = "dashed") +
  scale_colour_discrete(labels = c("Quebec", "Tax removed")) +
  labs(x = NULL, y = "Average price (cents per litre)", colour = NULL)
```



Comment on the results. Describe how the two lines move before April 2025, and what happens to the gap between them in April 2025. How does the change in the gap compare with the size of the carbon tax you found in Question 1?

Answer:

Before April 2025 the two lines move closely together. Both rise in January and fall back in March, and the gap between them stays within about 3 cents, between −2.9 cents (December) and +3.2 cents (March). In April the average price in the cities where the tax was removed falls by 18.4 cents, while the average in the two Quebec cities rises by 0.75 cents, so the gap changes by −19.1 cents in a single month. That is almost exactly the average carbon tax per litre in those 13 cities from Question 1, 19.4 cents. No other month-to-month change in the gap is larger than 4 cents.

After April the gap stays wide, between −11 and −16 cents, although it narrows by a few cents over the summer.

Question 3 (5 marks)

The simplest way to measure how much of the tax cut reached drivers is to compare the price just before the tax was removed with the price just after.

For each city, calculate the change in the price of gasoline from March 2025 to April 2025 (the April price minus the March price, so a fall is negative). Store the result in a data frame called `q3`, with one row per city and columns called `city`, `removed`, `carbon` (the tax in March) and `change`. Present `q3` as a table.

Then calculate the **pass-through rate** for each of the 13 cities where the tax was removed: the fall in price divided by the tax removed, or `-change / carbon`. A pass-through rate of 1 means the price fell by exactly the amount of the tax. Store the average of the 13 rates, as a single number, in `pt_naive`.

Hints.

- Keep only March and April 2025 with `filter()`. A row cannot be in both months at once, so combine the two conditions with `|` (or), not `&`. See "`filter()` — keep some rows" in the R tutorial.
- Then group by `city` and `removed` (grouping by `removed` as well keeps it as a column in `q3`), and use `summarise()` to create `carbon` and `change`.
- Inside `summarise()`, square brackets pick out the values in the rows that meet a condition: `gas[month == "2025-04"]` is the city's April price. See the income gap example under "`group_by()` — do it separately for each group" in the R tutorial.
- `carbon` is positive in March and 0 in April, so `max(carbon)` gives the tax in March.
- `datasummary_df()` turns a data frame into a table, and `fmt = 2` shows the numbers in it to two decimal places. See "Tables of data" in the R tutorial.

- For `pt_naive`, save the rows of `q3` where `removed == 1` as a new data frame. Calculate the 13 rates from its columns, which you can pick out with `$`, and take their `mean()`. See "Columns are vectors" in the R tutorial.

```
q3 <- gas |>
  filter(month == "2025-03" | month == "2025-04") |>
  group_by(city, removed) |>
  summarise(
    carbon = max(carbon),
    change = gas[month == "2025-04"] - gas[month == "2025-03"],
    .groups = "drop"
  )

datasummary_df(q3, fmt = 2)
```

city	removed	carbon	change
Calgary	1.00	18.49	−17.70
Charlottetown	1.00	20.25	−16.40
Edmonton	1.00	18.49	−18.60
Halifax	1.00	20.25	−17.90
Montreal	0.00	0.00	0.30
Ottawa	1.00	19.90	−21.20
Quebec City	0.00	0.00	1.20
Regina	1.00	18.49	−15.00
Saint John	1.00	20.25	−19.10
Saskatoon	1.00	18.49	−18.30
St. John's	1.00	20.25	−19.30
Thunder Bay	1.00	19.90	−19.60
Toronto	1.00	19.90	−20.20
Vancouver	1.00	18.49	−17.20
Victoria	1.00	18.49	−18.10

```
q3_removed <- q3 |> filter(removed == 1)
pt_naive <- mean(-q3_removed$change / q3_removed$carbon)
pt_naive
```

```
[1] 0.9483443
```

Comment on the results. Report `pt_naive` and explain what it means. What happened to the price in Montreal and Quebec City over the same month, when their taxes did not change? What does that tell you about using the before-and-after change to measure pass-through?

Answer:

`pt_naive` is 0.948: on average, the price fell by 94.8% of the carbon tax removed between March and April 2025. The rates range from about 0.81 (Charlottetown and Regina) to 1.07 (Ottawa); Toronto's price fell by 20.2 cents against a tax of 19.9 cents (1.02).

Over the same month, the price rose by 0.3 cents in Montreal and 1.2 cents in Quebec City, even though nothing changed in their taxes. The before-and-after change assumes that nothing other than the tax affected prices between March and April. The Quebec cities show that something else did: prices would probably have risen by a little under a cent everywhere without the tax change. (Crude oil actually got cheaper in April 2025, by roughly 5 to 6 cents a litre, but other costs, such as refining margins, which usually rise in the spring, more than offset it.) So the before-and-after comparison slightly *understates* pass-through here. The bias happens to be small this month, but it need not be: from December to January, prices rose by 4.7 cents in Quebec and 6.5 cents elsewhere, with no tax change at all. And without a comparison group we would not know how large the bias is.

Question 4 (5 marks)

The price of gasoline changes from month to month for many reasons other than taxes, such as the price of crude oil, the exchange rate and the season. A regression with **fixed effects** separates the effect of the tax from those other changes:

$$gas_{it} = \beta_0 + \beta_1 carbon_{it} + \text{city dummies} + \text{month dummies} + \epsilon_{it}$$

where i is a city and t is a month. The city dummies absorb differences between cities that stay the same over time, such as provincial fuel taxes and transport costs. The month dummies absorb changes that hit every city at the same time, such as a fall in the price of crude oil. What is left to estimate β_1 is how prices changed in the cities whose tax changed *compared with* the Quebec cities, whose tax did not. β_1 is the pass-through rate.

Because `city` and `month` are text, R creates the dummies for you when you put them in the formula.

(a) Keep only March and April 2025, and estimate `lm(gas ~ carbon + city + month, data = ...)` on those two months. Store the regression as `q4_model`, and store the coefficient on `carbon`, as a single number, in `pt_fe`.

Hints.

- Save the March and April rows as a new data frame with `filter()`, as in Question 3, and use it as the `data`.
- `coef()` gives all the coefficients of a regression. To keep just one, put its name in square brackets after it. See "Categorical predictors" in the R tutorial.

```
marapr   <- gas |> filter(month == "2025-03" | month == "2025-04")
q4_model <- lm(gas ~ carbon + city + month, data = marapr)
pt_fe    <- coef(q4_model)["carbon"]
pt_fe
```

```
   carbon
0.9848186
```

(b) Estimate the same regression using all 12 months, and store it as `q4_all`. Present both regressions in one table, one in each column, without the city and month dummies.

Hints.

- `modelsummary()` takes a list of regressions, and the names you give them become the column headings. See "Presenting results" in the R tutorial.
- `coef_omit = "city|month"` leaves the city and month dummies out of the table, which would otherwise run to 25 rows, and `gof_map = c("nobs", "r.squared")` keeps just the number of observations and the R^2 at the bottom.

```
q4_all <- lm(gas ~ carbon + city + month, data = gas)

modelsummary(list("March-April 2025" = q4_model, "All 12 months" = q4_all),
             coef_omit = "city|month", gof_map = c("nobs", "r.squared"))
```

	March-April 2025	All 12 months
(Intercept)	129.588	129.786
	(1.225)	(1.276)
carbon	0.985	0.745
	(0.056)	(0.052)
Num.Obs.	30	180
R2	0.998	0.974

Comment on the results. Interpret the coefficient on `carbon` in the March-April regression, and compare it with `pt_naive` from Question 3. Why are they different? Then explain why

the estimate from all 12 months is different again. Look at your graph from Question 2: does the gap between the two lines stay the same from October 2024 to March 2025? Which of the two estimates do you find more convincing, and why?

Answer:

The coefficient on `carbon` in the March-April regression is 0.985: compared with the change in the Quebec cities over the same month, removing one cent of carbon tax lowered the price by 0.985 cents. The standard error is 0.056, so the 95% confidence interval runs from about 0.86 to 1.10, which includes 1: we cannot reject that the tax was passed through to drivers in full.

It is higher than `pt_naive` (0.948) because the month dummy absorbs the change that hit every city, which the Quebec cities show was a small *rise* in price. The regression measures each city's fall against that rise, rather than against no change at all, and so attributes a slightly larger fall to the tax. (It also weights the cities differently from a simple average of the 13 rates, but that matters less.)

The estimate from all 12 months is lower, 0.745, because it compares the average gap between the two groups over the six months before the change with the average gap over the six months after. That is the right comparison only if the gap would otherwise have stayed constant, and the graph in Question 2 shows it did not. Before April it drifted up, from about −3 cents in December to +3 cents in March, and after April it narrowed again, from −16 cents in April to −11 cents in August. These movements have nothing to do with the tax. They reflect things like regional differences in refinery supply, seasonal patterns in prices and other policy changes (Quebec, for example, ended its regulated minimum price for gasoline in June 2025). But the 12-month regression counts them as part of the tax effect, and here they work against it: the pre-period average gap is lower than the March gap, and the post-period gap is narrower than the April gap.

The March-April estimate is more convincing. Over one month there is little time for the two groups to drift apart for other reasons, and the 19-cent change in the gap in April dwarfs every other month-to-month change, none of which is larger than 4 cents. Either regression measures pass-through only if, without the tax change, prices outside Quebec would have moved the same way as prices in Quebec. That assumption is much easier to believe for one month than for six. Even so, month-to-month changes in the gap of a few cents mean the one-month estimate could be off by 0.1 or more.

Question 5 (5 marks)

In class we found that a unit tax t raises the price consumers pay by

$$\Delta P_c = \frac{\varepsilon_s}{\varepsilon_s + |\varepsilon_d|}\, t$$

so the pass-through rate is $\theta = \varepsilon_s/(\varepsilon_s + |\varepsilon_d|)$.

(a) Rearrange this equation to express the elasticity of supply, ε_s, in terms of θ and $|\varepsilon_d|$, and show your steps under **Answer:** below. Then take the short-run elasticity of demand for gasoline to be -0.2, and calculate the elasticity of supply implied by `pt_fe`, your estimate from Question 4(a). Store it, as a single number, in `eps_s`.

Hint. Calculate `eps_s` from `pt_fe` itself, not from a number copied out of your table. The table rounds `pt_fe`, and because `pt_fe` is close to 1, even a small rounding error changes `eps_s` a lot.

```
eps_s <- pt_fe * 0.2 / (1 - pt_fe)
eps_s
```

```
  carbon
12.97401
```

```
# For comparison, with the all-months estimate from Question 4(b):
pt_all <- coef(q4_all)["carbon"]
pt_all * 0.2 / (1 - pt_all)
```

```
   carbon
0.5857918
```

(b) Comment on the results. Who bore the economic burden of the carbon tax at the pump? In your answer, consider:

- the value of `eps_s`, and why the supply of gasoline to a single province might be this elastic. How would `eps_s` change if you used the all-months estimate from Question 4(b) instead?
- the charge was paid to the government by fuel distributors, not by drivers. Does that change who bore it?
- the Canada Carbon Rebate ended along with the tax. A household's rebate depended on its province, its size and whether it lived in a rural area, but not on how much fuel it bought, while the tax it paid rose with every litre. Given your answer about who bore the tax, which households were better off when the tax and the rebate ended together, and which were worse off?

Answer:

(a) Starting from $\theta = \varepsilon_s/(\varepsilon_s + |\varepsilon_d|)$:

$$\theta\varepsilon_s + \theta|\varepsilon_d| = \varepsilon_s \quad \Rightarrow \quad \theta|\varepsilon_d| = \varepsilon_s(1-\theta) \quad \Rightarrow \quad \varepsilon_s = \frac{\theta|\varepsilon_d|}{1-\theta}$$

With $\theta = 0.985$ and $|\varepsilon_d| = 0.2$, the implied elasticity of supply is `eps_s` $= 12.97$.

(b) Drivers bore almost all of the carbon tax. With pass-through close to 1, removing the tax lowered the price drivers pay by nearly the full amount of the tax, so the price sellers receive, net of tax, hardly changed. An elasticity of supply of about 13, against an elasticity of demand of -0.2, means supply is roughly 65 times as elastic as demand, so the formula puts 98.5% of the burden on consumers.

Supply to a single province is this elastic because gasoline is sold in a North American market. Refiners and wholesalers can ship gasoline to another province or state, so they will not accept a lower net price in one province than they can get elsewhere, and one province is too small a part of the market to move that price. The estimate is very sensitive to θ when θ is close to 1: with the all-months estimate of 0.745, `eps_s` falls to 0.59, which would put about a quarter of the burden on sellers. Since the March-April confidence interval includes 1, the data are also consistent with perfectly elastic supply, where consumers bear the whole tax.

That distributors, not drivers, paid the charge to the government makes no difference. As we saw in class, statutory incidence does not determine economic incidence. Distributors remitted the tax but raised their prices by almost the full amount, so the burden fell on drivers. The result would have been the same if drivers had paid the charge directly.

Because drivers bore the tax, each household's tax rose with the fuel it bought, while its rebate did not. When both ended, households that bought a lot of fuel gained: their savings at the pump exceeded the rebate they lost. That includes households with long commutes or several vehicles, which are more often rural and higher-income households. Households that bought little fuel lost more rebate than they saved at the pump. That includes households without a car, those who rely on transit, and on average lower-income households. So removing the tax and the rebate together made the system less progressive: the rebate had been worth more, relative to income, to lower-income households.

Use of generative AI

If you used a generative AI tool for any part of this assignment, name the tool and describe what you used it for. If you did not use one, say so.

Not applicable: this is the answer key. Students must say whether and how they used generative AI.

Before you submit

This chunk lists any required objects that are still missing. It should say "All required objects found" before you submit.

```
All required objects found
```